

Notes on QFT

PROBABILITY FLOOR

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Source: <https://github.com/probability569/qft-notes-schwartz>

“note to self, do this or im the biggest retard”

feyman likely

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Preface

todo

1 Mathematical Prerequisite and Notations

abc abc

$$F = ma \quad (1.1)$$

eqn sec thm sol pbm plm lem chp

1.1 Linear Algebra Review

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2 Classical Field Theory

2.1 Classical Field Theory

- The idea behind **classical field theory** is to utilize the idea of Lagrangians that will be presented in *Lagrangians* and in *Classical Symmetries* to obtain a “toy theory” without the quantum effects that will later appear such as Loops in diagrams, etc. In practice, this means that it is essentially the *Lagrangian Formalism* of Classical Mechanics where *position* and *momentum* coordinates (p, q) are replaced by fields (ϕ, π) , as seen before in *Lagrangians* without having been explicitly stated.
- Classical Field Theory, much like *Classical Mechanics*, functions as the unquantized equivalent to Quantum Field Theory. Quantization does not require much to achieve as simply putting quantization conditions on the operators being used achieves it with relative ease.

Example 2.1.1 (Klein Gordon Equation).

- A very standard equation for a field in Classical Field theory is the *Klein Gordon Equation*. (This, in fact, describes a *free* field in which there are no interactions, allowing for a very good “toy theory” to use and potentially apply later).

$$(\square + m^2)\phi = 0$$

or

$$(\partial^2 + m^2)\phi = 0$$

is the *Klein Gordon Equation*.

3 QCD Feynman Rules

3.1 QCD Feynman Rules

- Now, we have the full lagrangian for QCD (the F-P) one allowing us to read off feynman rules

- $$\mathcal{L} = -\frac{1}{4}(F_{\mu\nu}^a)^2 - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 +$$

$$(\partial_\mu \bar{c}^a)(\delta^{ac}\partial_\mu + gf^{abc}A_\mu^b)c^c + \bar{\psi}_i(i\not{\partial}\delta_{ij} + gA_\mu^a T_{ij}^a - m\delta_{ij})\psi_j +$$

$$[(\delta_{ki}\partial_\mu - igA_\mu^a T_{ki}^a)\phi_i]^\dagger[(\delta_{kj}\partial_\mu - igA_\mu^a T_{kj}^a)\phi_j] - m^2\phi_i^\dagger\phi_i$$

- The kinetic terms for the lagrangian are

$$\mathcal{L}_{\text{kin}} = -\frac{1}{4}(\partial_\mu A_\mu^a - \partial_\nu A_\nu^a)^2 -$$

$$\frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + \bar{\psi}_i(i\not{\partial} - m)\psi_i - \phi_i^*(\square + m^2)\phi_i - \bar{c}^a\square c^a$$

- The rules are as follows:

$$\mu; a \text{ --- } \nu; b = \frac{-g^{\mu\nu} + (1 - \xi)\frac{p^\mu p^\nu}{p^2}}{p^2} \delta^{ab}$$

- For ghosts,

$$\text{---} \Rightarrow \text{---} = \frac{\delta^{ab}}{p^2 + i\epsilon}$$

- For colored fermions,

$$i \longrightarrow j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon}$$

- For colored scalars,

$$i \text{ ---} \Rightarrow \text{---} j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon}$$

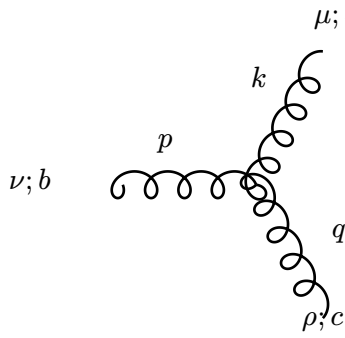
- The interaction terms (which are where this gets interesting) are the following:

$$\mathcal{L}_{int} = -gf^{abc}(\partial_\mu A_\nu^a)A_\mu^b A_\nu^c - \frac{1}{4}g^2(f^{eab}A_\mu^a A_\nu^b)(f^{ecd}A_\mu^c A_\nu^d)$$

$$+ gf^{abc}(\partial_\mu \bar{c}^a)A_\mu^b c^c + gA_\mu^a \bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j$$

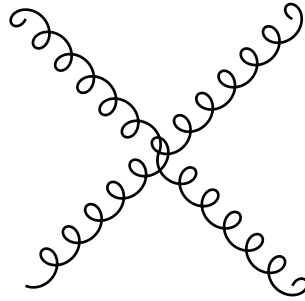
$$+ igA_\mu^a T_{ij}^a (\phi_i \overleftrightarrow{\partial}^\mu \phi_j) + g^2 \phi_i^* A_\mu^a T_{ik}^a T_{kj}^b A_\mu^b \phi_j$$

- Trips g:

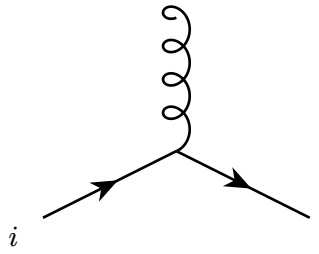


$$= gA^{abc}[g^{\mu\nu}(k-p)^\mu + g^{\nu\rho}(p-q)^\mu + g^{\rho\mu}(q-k)^\nu]$$

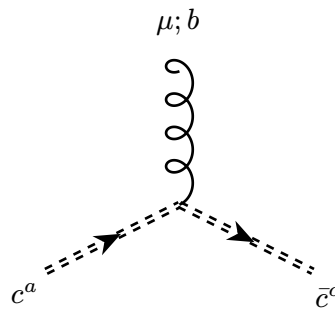
- Quads g:



$$= -ig^2[f^{abe}f^{cde}(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\sigma}g^{\nu\rho}) + f^{ace}f^{bde}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\sigma}g^{\nu\rho}) + f^{ade}f^{bce}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma})]$$



$$= ig\gamma^\mu T_{ij}^a$$



$$= -gf^{abc}p^\mu$$

$= ig(k^\mu + q^\mu)T_{ij}^a$

$= ig^2 T_{ik}^a T_{kj}^b g^{\mu\nu}$

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